

# Content-Rich Reflected Fibers in a Primitive Möbius Tail:

## Long-Fiber Pruning, a Sparse Incidence Wedge, and the Primitive-Core Barrier

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### Abstract

Let  $P_{m,T}$  be the calibrated divisor prefix at the upper cutoff of a selected primitive opened-row packet, and let  $U_0 \asymp X$  contain all positive targets on its physical support. The preceding conductor argument closes the truncated quadratic packet at  $T$ , but does not estimate the ultra-long complement. We begin from the exact calibrated identity

$$P_{m,U_0} - P_{m,T} = C_{m;T,U_0}, \quad C_{m;T,U_0}(j) = \sum_{\substack{T < w \leq U_0 \\ w \mid mj+h}} -\mu(w) \log w.$$

It avoids a scalar-tail replacement whose available pointwise bounds do not by themselves imply arbitrary logarithmic saving. Expanding the joint cutoff difference leaves three raw ultra channels and two row-drift legs; the latter are power-small from the existing  $L^{-1}$  row factor.

Reflect a long divisor  $w$  to  $r = (mj + h)/w$ . For distinct primitive rows  $m, n$ , dyadic complement variables

$$r \asymp A, \quad s \asymp B, \quad (r, s) \asymp G$$

select, when compatible, one orbit class modulo  $[r, s]$ . We prove the fixed-row incidence estimate

$$\mathcal{N}_{m,n}(A, B, G) \ll_{\varepsilon, h} X^\varepsilon \min \left\{ J, \frac{AB}{G^2} + \frac{J}{G} \right\}.$$

The first term is the pointwise divisor bound. For the second, each exact common content divides  $h(m - n)$ ; after writing  $r = dr_1$ ,  $s = ds_1$ , the number of coprime primitive pairs is  $O(AB/d^2)$  and their orbit occupancy is  $O(1 + Jd/(AB))$ . This argument neither removes the Möbius signs nor assumes that complement and quotient contents are coprime.

The actual opened-row coefficients have one-row  $\ell^1$  mass  $O(Q)$ . Consequently every physical mixed or both-ultra dyadic block satisfies

$$|\mathcal{S}(A, B, G)| \ll_\varepsilon Q^2 X^\varepsilon \min \left\{ J, \frac{AB}{G^2} + \frac{J}{G} \right\}.$$

Writing  $q = [r, s]$  and  $g = (r, s)$ , this gives the unconditional content-rich sparse wedge

$$q \geq J, \quad \frac{q}{g} \leq JX^{-\eta} \implies \mathcal{S}_{\text{rich}} \ll_\varepsilon XQX^{-\eta+\varepsilon}.$$

Here  $q/g = (r/g)(s/g)$  is the primitive lcm after the shared content is removed. On the complementary long-fiber side  $q \leq Y \leq J$ , row residue energy also gives

$$\mathcal{S}_{q \leq Y, (r,s) > D} \ll_\varepsilon X^\varepsilon \left\{ \frac{XQ}{D} \log^2(2Y) + X \log^3(2Y) \right\}.$$

Thus every polynomially large shared-content sector is power-small on prescribed long-fiber scales as well.

At the high- $\beta$  sample  $\beta = 267/400$ ,  $T = X^{193/500+o(1)}$ , the block  $A = B = X^{1/4}$ ,  $G = X^{1/10}$  is genuinely sparse and has the exact incidence margin  $13/400$ .

This closes a nonempty exponent region of the ultra-long geometry without a Chowla or Elliott input. It does not close the primitive sparse core, the dense long-fiber region, the complete residual, positivity, a Hardy–Littlewood asymptotic, twin primes, or the sieve parity problem.

**Keywords:** primitive Möbius tail; divisor reflection; sparse fiber; incidence geometry; primitive lcm; sieve parity.

**MSC 2020:** Primary 11N35, 11N37; Secondary 11B25, 11D45.

## 1 Introduction and claim ledger

The primitive residual program separates a fixed-shift prime-pair correlation into opened row packets before confronting the remaining Möbius signs. TPC-23 identified the exact affine fibers produced by reflecting divisors beyond a dynamic cutoff [3]. TPC-26 then treated a finite annular difference jointly, and TPC-28 closed a source-compatible selected square through an upper cutoff  $T$ , even though its base already exceeds the Poisson length [5, 6]. The resulting statement is a rigorous truncated-packet estimate. The terms with a divisor larger than  $T$  remain.

The present paper removes a new, explicitly defined part of that remainder. It does not ask for cancellation on any individual Möbius fiber. Instead it exploits the common content of the two reflected divisors. If

$$r \mid mj + h, \quad s \mid nj + h,$$

then  $(r, s)$  divides  $h(m - n)$ . In the primitive physical packet it is coprime to  $h$ , hence divides the nonzero row difference. A large shared content therefore has two simultaneous effects: it restricts the row pair and reduces the number of primitive modulus pairs. These effects are strong enough to close two nonempty scale regions without using the signs of  $\mu$ .

### 1.1 The new results

The proof has four layers.

- (i) We retain the calibrated prefix and take its exact cutoff difference. If  $U_0$  contains all physical targets, then

$$P_{m,U_0} - P_{m,T} = C_{m;T,U_0} \tag{1}$$

with no scalar centering error. The joint difference has three raw ultra channels and two power-small row-drift legs.

- (ii) For a fixed off-diagonal row pair and dyadic divisor scales  $A, B, G$ , we prove

$$\mathcal{N}_{m,n}(A, B, G) \ll_{\varepsilon,h} X^\varepsilon \min \left\{ J, \frac{AB}{G^2} + \frac{J}{G} \right\}. \tag{2}$$

The estimate is a count with multiplicity. It therefore applies after inserting arbitrary signs and all physical support selectors.

- (iii) The opened-row  $\ell^1$  provenance converts (2) into a physical block bound. In the sparse range  $q = [r, s] \geq J$ , every fixed  $\eta > 0$  gives

$$\frac{q}{(r, s)} \leq JX^{-\eta} \implies \mathcal{S}_{\text{rich}} \ll_\varepsilon XQX^{-\eta+\varepsilon}. \tag{3}$$

Thus a short fiber is useful only after the number of primitive modulus pairs is also controlled.

- (iv) In the long-fiber region  $q \leq Y \leq J$ , an independent row-residue argument removes every block with common content  $d = (r, s) > D$ :

$$\mathcal{S}_{q \leq Y, d > D} \ll_{\varepsilon} X^{\varepsilon} \left\{ \frac{XQ}{D} \log^2(2Y) + X \log^3(2Y) \right\}. \quad (4)$$

For  $D \geq X^{\eta}$  this has a fixed-power saving relative to  $XQ$ .

These two closure regions share the same mechanism but cover different fiber lengths. The long theorem uses the  $J/q$  occupancy and row residue energy. The sparse theorem keeps the unavoidable  $+1$  in the occupancy and compensates for it by requiring the primitive lcm  $q/(r, s)$  to be short.

At the TPC-28 high- $\beta$  sample,

$$\beta = \frac{267}{400}, \quad J = X^{133/400+o(1)}, \quad T = X^{193/500+o(1)},$$

the dyadic choice

$$A = B = X^{1/4}, \quad G = X^{1/10}$$

has lcm exponent  $2/5$  and primitive-lcm exponent  $3/10$ . It is sparse by  $27/400$  and the new incidence saving is exactly  $13/400$ . This  $13/400$  is unrelated to the smaller  $13/1600$  conductor margin that closed the TPC-28 base.

## 1.2 What remains

The theorem does not estimate every short fiber. When  $(r, s) = 1$  and  $[r, s] \geq J$ , the exponent delivered by (2) is exactly the natural exponent. A new signed estimate after summing the growing family is still needed. Nor does (4) remove the long fibers with small content. Their affine slopes grow polynomially with the row scale, whereas the fixed-form logarithmic Elliott theorem has no uniform rate that can be summed over the physical family [1].

No conclusion below concerns positivity, a prime-pair main term, or infinitely many twin primes. Even a future closure of every ultra channel would still have to be reassembled with the full TPC-18 endpoint and block ledger before the residual-dispersion target was reached [2].

## 1.3 Organization

Section 2 fixes the physical packet. Section 3 proves the exact joint cutoff identity and removes the row drifts. Section 4 derives the reflected fibers and the incidence bound. Section 5 proves the physical sparse theorem, while section 6 proves the long-fiber companion theorem. Section 7 gives the rational sample. Section 8 identifies the remaining signed gate, and section 9 records the certificate and exact scope.

## 2 The calibrated physical packet

Fix  $h \in \mathbb{Z} \setminus \{0\}$  and put  $H = \text{rad } |h|$ . We retain one primitive opened- $d$  packet from TPC-25–TPC-28. Its scales satisfy

$$QJ \asymp X, \quad Q = X^{\beta+o(1)}, \quad J = X^{1-\beta+o(1)}, \quad 0 < \beta < 1. \quad (5)$$

The inherited sieve base satisfies  $1 \leq R < L/2$ . The row labels are  $\alpha = (\ell_{\alpha}, d_{\alpha})$ , where each  $\ell_{\alpha} \asymp L$  is prime,  $d_{\alpha}$  is squarefree, and  $d_{\alpha} \leq R^{1/2}$ . They have distinct row integers

$$m_{\alpha} = \ell_{\alpha} d_{\alpha} \asymp Q, \quad L \geq X^{\lambda_0} \quad (6)$$

for a fixed  $\lambda_0 > 0$ . The size separation in the opened packet makes  $m_\alpha$  determine  $\alpha$  uniquely. We impose the primitive orbit conditions

$$(m_\alpha, H) = 1, \quad (j, H) = 1. \quad (7)$$

Consequently, for

$$N_\alpha(j) = m_\alpha j + h, \quad (8)$$

every divisor of a positive target is coprime to  $m_\alpha H$ .

The two opened-row coefficients have the form described in TPC-25 [4] and satisfy

$$\sum_{\alpha} |\gamma_{\alpha}^{(i)}| \ll_{\mathcal{D}} Q, \quad (9)$$

$$\sum_{\alpha} |\gamma_{\alpha}^{(i)}|^2 \ll_{\mathcal{D}} Q \log(2L), \quad i = 1, 2. \quad (10)$$

Here and below  $\mathcal{D}$  denotes the fixed smooth, residue, source, and support data. The row-pair mask satisfies

$$|\mathbf{m}(\alpha, \gamma)| \leq 1, \quad \mathbf{m}(\alpha, \gamma) = 0 \quad \text{if } m_\alpha = m_\gamma, \quad (11)$$

and is independent of every divisor variable. The fixed-period factor  $\Xi_{\alpha, \gamma}(j)$  and the smooth weight  $\mathcal{W}_{\alpha, \gamma}(j/J)$  are uniformly bounded. Their joint support is contained in an interval of  $O_{\mathcal{D}}(J)$  consecutive integers and satisfies

$$c_{\mathcal{D}} X \leq N_\alpha(j) \leq C_{\mathcal{D}} X \quad (12)$$

whenever it contributes. In particular every target below is positive and nonzero. Every  $j$ -sum below retains the primitive-orbit condition (7) and this joint physical support.

## 2.1 Divisor coefficients and prefixes

Let

$$a(w) = -\mu(w) \log w, \quad b_R(w) = a(w) - \lambda'_R(w), \quad (13)$$

where  $\lambda'_R(w) = 0$  for  $w > R$ . In the polynomial support ranges used here,

$$|a(w)| + |b_R(w)| \ll_{\mathcal{D}} (\log(2X))^3. \quad (14)$$

For  $m = \ell d$ , the calibrated row drift is

$$\delta_H(m) = \frac{H}{\varphi(H)} \frac{d}{\varphi(d)(\ell - 1)} \ll_{\varepsilon, \mathcal{D}} L^{-1} X^\varepsilon. \quad (15)$$

For  $R \leq W$ , put

$$\mathbf{A}_{m, W}(j) := \sum_{\substack{w \leq W \\ w | m j + h}} b_R(w), \quad (16)$$

$$P_{m, W}(j) := \mathbf{A}_{m, W}(j) - \delta_H(m). \quad (17)$$

The elementary divisor bound gives, uniformly on the physical support,

$$|\mathbf{A}_{m, W}(j)| \ll_{\varepsilon, \mathcal{D}} X^\varepsilon. \quad (18)$$

Choose an integer horizon  $U_0 \asymp_{\mathcal{D}} X$  larger than every positive target in (12), and fix

$$R \leq T < U_0. \quad (19)$$

The physical quadratic packet at cutoff  $W$  is

$$\begin{aligned} \mathcal{Q}_W^{\text{phys}} := & \sum_{\alpha, \gamma} \gamma_{\alpha}^{(1)} \gamma_{\gamma}^{(2)} \mathfrak{m}(\alpha, \gamma) \sum_j \Xi_{\alpha, \gamma}(j) \mathcal{W}_{\alpha, \gamma}(j/J) \\ & \times P_{m_{\alpha}, W}(j) P_{m_{\gamma}, W}(j). \end{aligned} \quad (20)$$

Its natural absolute scale is

$$JQ^2 \asymp XQ. \quad (21)$$

TPC-28 closes (20) at the selected upper cutoff  $T$  under its additional conductor interfaces. The estimates proved below concern specified regions of the exact difference  $\mathcal{Q}_{U_0}^{\text{phys}} - \mathcal{Q}_T^{\text{phys}}$  and use only the physical-space interfaces displayed in this section.

### 3 The exact calibrated ultra difference

Define the raw ultra-long increment

$$C_{m, T, U_0}(j) := \sum_{\substack{T < w \leq U_0 \\ w | m\tilde{j} + h}} a(w). \quad (22)$$

We often abbreviate it to  $C_m(j)$ .

**Proposition 3.1** (Exact calibrated cutoff identity). *For every physical row and orbit point,*

$$\boxed{P_{m, U_0}(j) - P_{m, T}(j) = C_{m, T, U_0}(j)}. \quad (23)$$

Consequently, for two rows  $m, n$ ,

$$\begin{aligned} P_{m, U_0} P_{n, U_0} - P_{m, T} P_{n, T} \\ = A_{m, T} C_n + C_m A_{n, T} + C_m C_n - \delta_H(m) C_n - \delta_H(n) C_m. \end{aligned} \quad (24)$$

All identities are pointwise in  $j$ .

*Proof.* The drift  $\delta_H(m)$  is independent of the cutoff and cancels in the difference. If  $w > T \geq R$ , then  $\lambda'_R(w) = 0$  and hence  $b_R(w) = a(w)$ . Subtracting the two finite divisor prefixes proves (23). Now write  $P_{m, T} = A_{m, T} - \delta_H(m)$  and expand

$$(P_{m, T} + C_m)(P_{n, T} + C_n) - P_{m, T} P_{n, T}.$$

This gives (24). □

The first three terms on the right of (24) will be called the two mixed raw channels and the both-ultra raw channel. They are not asserted to be small separately on their full supports. Absolute estimates will be used only after restricting them to the content-rich regions defined later.

**Proposition 3.2** (The two drift legs are power-small). *Let  $\mathcal{D}_{T, U_0}^{\text{phys}}$  denote the contribution of the last two terms in (24) to  $\mathcal{Q}_{U_0}^{\text{phys}} - \mathcal{Q}_T^{\text{phys}}$ . Then, for every fixed  $\varepsilon > 0$ ,*

$$\boxed{|\mathcal{D}_{T, U_0}^{\text{phys}}| \ll_{\varepsilon, \mathcal{D}} XQ L^{-1} X^{\varepsilon}}. \quad (25)$$

Thus it has a fixed-power saving when  $\varepsilon < \lambda_0$ .

*Proof.* On the physical support,

$$|C_m(j)| \leq \tau(N_m(j)) \log(2X) \ll_{\varepsilon, \mathscr{D}} X^\varepsilon. \quad (26)$$

Delete the bounded mask, periodic factor, and smooth weight. Use (15) on the drift row, (9) on both row families, and the  $O(J)$  orbit support. Each leg is

$$\ll J(QL^{-1}X^\varepsilon)QX^\varepsilon \ll XQL^{-1}X^{3\varepsilon},$$

after  $JQ \asymp X$ . Rename the auxiliary exponent.  $\square$

### 3.1 Why the exact difference matters

One may instead center the ultra divisor sum by subtracting its scalar density

$$\tau_{T,U_0}(m) = \sum_{\substack{T < w \leq U_0 \\ (w, m\bar{H})=1}} \frac{a(w)}{w}. \quad (27)$$

The pointwise centered-tail interface considered in TPC-23 [3] supplies majorants of the shape

$$\tau_{T,U_0}(m) \ll_{A,\varepsilon} X^\varepsilon (\log X)^{-A}, \quad C_m(j) \ll_\varepsilon X^\varepsilon. \quad (28)$$

These displayed majorants alone do not imply  $O_B((\log X)^{-B})$  for every  $B$ : a fixed power  $X^\varepsilon$  cannot be absorbed by choosing a fixed logarithmic order. An additional average tail norm would repair such a replacement, but it is not needed here. The exact identity (23) contains no scalar tail and therefore avoids this interface altogether. We do not import that pointwise centered-tail replacement in any theorem below.

## 4 Reflected fibers and a sparse incidence theorem

### 4.1 Hyperbola reflection

For a positive target  $N_m(j) = mj + h$ , divisor reflection in (22) gives the exact identity

$$C_m(j) = \sum_{\substack{r|N_m(j) \\ r < N_m(j)/T}} a\left(\frac{N_m(j)}{r}\right). \quad (29)$$

The reflected variable  $r$  is not required to be squarefree. Only the quotient coefficient  $a(N_m/r)$  vanishes off squarefree quotients.

The both-ultra channel contains two variables of the form (29). In a mixed channel, one variable is an original prefix divisor  $u \leq T$  and the other is a reflected complement variable. Both situations reduce to the same pair of divisibility conditions

$$r \mid mj + h, \quad s \mid nj + h. \quad (30)$$

Support restrictions and coefficient types will be restored only after the common incidence count is proved.

**Lemma 4.1** (Compatibility and the reflected determinant). *Let  $m \neq n$  be primitive physical rows, and let  $r, s$  satisfy (30) for some primitive orbit point. Put*

$$d = (r, s), \quad q = [r, s] = \frac{rs}{d}. \quad (31)$$

Then the two congruences in (30) are compatible if and only if

$$d \mid h(m - n). \quad (32)$$

When compatible, they select a unique class  $j_0 \pmod{q}$ . Writing  $j = j_0 + qk$ , the two quotient forms

$$U(k) = \frac{m(j_0 + qk) + h}{r} = A_1 k + B_1, \quad (33)$$

$$V(k) = \frac{n(j_0 + qk) + h}{s} = A_2 k + B_2 \quad (34)$$

have integer coefficients and satisfy

$$\boxed{A_1 B_2 - A_2 B_1 = \frac{h(m - n)}{d} \neq 0.} \quad (35)$$

Moreover  $(d, h) = 1$ , so compatibility implies  $d \mid m - n$ .

*Proof.* Target primitivity gives  $(r, m) = (s, n) = 1$ . Hence the two congruences select

$$j \equiv -h\bar{m} \pmod{r}, \quad j \equiv -h\bar{n} \pmod{s}.$$

Their reductions modulo  $d$  agree exactly when  $d \mid h(m - n)$ . The generalized Chinese remainder theorem then gives one class modulo  $q$ .

The affine coefficients are

$$A_1 = \frac{mq}{r}, \quad B_1 = \frac{mj_0 + h}{r}, \quad A_2 = \frac{nq}{s}, \quad B_2 = \frac{nj_0 + h}{s}.$$

Their determinant equals

$$\frac{qh(m - n)}{rs} = \frac{h(m - n)}{d}.$$

It is nonzero by the off-diagonal mask. Finally,  $d$  divides both positive targets. Each target is coprime to  $H = \text{rad } |h|$ , so  $(d, h) = 1$ ; cancel  $h$  in (32).  $\square$

Equivalently, the quotient variables lie on the exact surface

$$nrU - msV = h(n - m). \quad (36)$$

No coprimality between  $d$  and either quotient is asserted. Such an assumption would be false for nonsquarefree targets and is not used below.

## 4.2 A dyadic count with multiplicity

Let  $\mathcal{I} \subset \mathbb{Z}$  be any set contained in an interval of at most  $C_{\mathcal{Q}}J$  consecutive integers, and suppose throughout this section that

$$1 \leq mj + h, \quad nj + h \leq X^{C_{\mathcal{Q}}} \quad (j \in \mathcal{I}). \quad (37)$$

The physical target window (12) is the case used later. For  $A, B, G \geq 1$ , define

$$\begin{aligned} \mathcal{N}_{m,n}(A, B, G; \mathcal{I}) := \# \{ (j, r, s) : j \in \mathcal{I}, \quad A < r \leq 2A, \quad B < s \leq 2B, \\ G < (r, s) \leq 2G, \quad r \mid mj + h, \quad s \mid nj + h \}. \end{aligned} \quad (38)$$

This is a triple count, not merely the number of occupied orbit points. At the left endpoint of a dyadic decomposition, the  $G = 1$  cell is understood as  $1 \leq (r, s) \leq 2$ ; subsequent cells use the displayed half-open convention.

**Theorem 4.2** (Fixed-row sparse incidence). *Uniformly for distinct physical rows  $m, n$  and all  $A, B, G \geq 1$ ,*

$$\boxed{\mathcal{N}_{m,n}(A, B, G; \mathcal{I}) \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon \min \left\{ J, \frac{AB}{G^2} + \frac{J}{G} \right\}.} \quad (39)$$

*Empty dyadic cells are understood to contribute zero.*

*Proof.* First fix an exact common content  $d = (r, s)$ . By lemma 4.1, it must divide the nonzero integer  $h(m - n)$ . The number of possible exact  $d$  in the dyadic interval is therefore at most

$$\tau(|h(m - n)|) \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon. \quad (40)$$

Write

$$r = dr_1, \quad s = ds_1, \quad (r_1, s_1) = 1. \quad (41)$$

For a nonempty cell,  $d \leq 2A$  and  $d \leq 2B$ . Hence the number of candidate pairs at this fixed  $d$  is

$$\ll \left( \frac{A}{d} + 1 \right) \left( \frac{B}{d} + 1 \right) \ll \frac{AB}{d^2}. \quad (42)$$

Deleting the coprimality condition only enlarges the count.

Each compatible pair selects one residue class modulo  $q = rs/d$ . Since  $r > A$  and  $s > B$ , its occupancy in  $\mathcal{I}$  is

$$\ll_{\mathcal{D}} 1 + \frac{J}{q} \ll_{\mathcal{D}} 1 + \frac{Jd}{AB}. \quad (43)$$

Multiplying (42) and (43), using  $d \asymp G$  (also in the closed  $G = 1$  base cell), and then applying (40) gives

$$\mathcal{N}_{m,n}(A, B, G; \mathcal{I}) \ll X^\varepsilon \left( \frac{AB}{G^2} + \frac{J}{G} \right). \quad (44)$$

There is a second bound. For each fixed  $j$ , the numbers of possible  $r$  and  $s$  are at most  $\tau(N_m(j))$  and  $\tau(N_n(j))$ . The pointwise divisor bound and  $\#\mathcal{I} \ll J$  give

$$\mathcal{N}_{m,n}(A, B, G; \mathcal{I}) \ll_{\varepsilon, \mathcal{D}} JX^\varepsilon. \quad (45)$$

Taking the smaller of (44) and (45) proves the theorem.  $\square$

*Remark 4.3* (The indispensable  $+1$ ). When  $q \geq J$ , (43) is  $O(1)$ , not  $O(J/q)$ . Dropping this  $+1$  would incorrectly turn every sparse family into a saving. The term  $AB/G^2$  in (39) is precisely the total cost of those one-point occupancies.

## 5 Physical weighting and the content-rich sparse wedge

We now restore the physical coefficients. For one of the three raw channels in (24), let  $\kappa_{\alpha, \gamma}(j, r, s)$  include its two divisor coefficients, its prefix or reflected support restrictions, and any fixed residue selectors. Thus it is one of the types

$$b_R(r)a(N_\gamma(j)/s), \quad a(N_\alpha(j)/r)b_R(s), \quad a(N_\alpha(j)/r)a(N_\gamma(j)/s), \quad (46)$$

on the appropriate divisor support, and is zero otherwise. By (14),

$$|\kappa_{\alpha, \gamma}(j, r, s)| \ll_{\mathcal{D}} (\log(2X))^6. \quad (47)$$

Define the corresponding dyadic physical block

$$\mathcal{S}(A, B, G) := \sum_{\alpha, \gamma} \gamma_\alpha^{(1)} \gamma_\gamma^{(2)} \mathbf{m}(\alpha, \gamma) \sum_j \Xi_{\alpha, \gamma}(j) \mathcal{W}_{\alpha, \gamma}(j/J)$$



$$\times \sum_{\substack{A < r \leq 2A, \ B < s \leq 2B \\ G < (r,s) \leq 2G \\ r|N_\alpha(j), \ s|N_\gamma(j)}} \kappa_{\alpha,\gamma}(j, r, s). \quad (48)$$

**Theorem 5.1** (Weighted block estimate). *Every mixed or both-ultra block of the form (48) satisfies*

$$|\mathcal{S}(A, B, G)| \ll_{\varepsilon, h, \mathcal{D}} Q^2 X^\varepsilon \min \left\{ J, \frac{AB}{G^2} + \frac{J}{G} \right\}. \quad (49)$$

*No Poisson, Fourier, conductor, or Möbius-correlation theorem is used.*

*Proof.* Take absolute values in (48). The mask, periodic factor, and smooth weight are bounded. Absorb (47) into  $X^\varepsilon$ , and apply [theorem 4.2](#) to each off-diagonal row pair. Finally use

$$\sum_{\alpha, \gamma} |\gamma_\alpha^{(1)} \gamma_\gamma^{(2)}| \leq \|\gamma^{(1)}\|_1 \|\gamma^{(2)}\|_1 \ll_{\mathcal{D}} Q^2$$

from (9). □

## 5.1 Primitive lcm and the sparse theorem

For a divisor pair, put

$$q = [r, s], \quad g = (r, s), \quad q_{\text{prim}} := \frac{q}{g} = \frac{r}{(r, s)} \frac{s}{(r, s)}. \quad (50)$$

Thus  $q_{\text{prim}}$  is the product of the two coprime parts after the shared content is removed.

**Definition 5.2** (Content-rich sparse region). For fixed  $\eta > 0$ , the content-rich sparse region consists of the divisor pairs satisfying

$$q \geq J, \quad q_{\text{prim}} \leq JX^{-\eta}. \quad (51)$$

Dyadic endpoint constants may be changed by fixed factors; all inequalities retain their fixed exponent margin.

**Theorem 5.3** (Unconditional content-rich sparse closure). *Let  $\mathcal{S}_{\text{rich}}$  be the sum, over the three raw channels in (24), of all dyadic blocks restricted to (51). Then*

$$|\mathcal{S}_{\text{rich}}| \ll_{\varepsilon, \eta, h, \mathcal{D}} XQX^{-\eta+\varepsilon}. \quad (52)$$

*In particular it is fixed-power small for  $0 < \varepsilon < \eta$ .*

*Proof.* On a dyadic block,  $q \asymp AB/G$  and  $q_{\text{prim}} \asymp AB/G^2$ . The second condition in (51) gives

$$\frac{AB}{G^2} \ll JX^{-\eta}. \quad (53)$$

Dividing the sparse lower bound for  $q$  by the upper bound for  $q_{\text{prim}}$  gives

$$G \gg X^\eta, \quad \frac{J}{G} \ll JX^{-\eta}. \quad (54)$$

Insert (53) and (54) into (49). Each block is  $O(Q^2 JX^{-\eta+\varepsilon})$ . Since  $Q^2 J \asymp XQ$ , summing the  $O((\log X)^3)$  scale triples and the three channel types proves (52) after reducing the auxiliary exponent. □

## 5.2 The exponent polytope

Write

$$A = X^{a+o(1)}, \quad B = X^{b+o(1)}, \quad G = X^{c+o(1)}, \quad j = 1 - \beta. \quad (55)$$

The sparse condition is

$$a + b - c \geq j. \quad (56)$$

The weighted theorem gives the exact saving exponent available from this incidence method:

$$\boxed{\eta_{\text{sp}}(a, b, c) = (j - a - b + 2c)_+}. \quad (57)$$

Indeed, the exponent of the second entry in the minimum in (49) is  $\max\{a + b - 2c, j - c\}$ . Under (56), one has  $a + b - 2c \geq j - c$ , so the first term is the larger one and (57) follows. Therefore a fixed saving  $\eta_0$  is equivalent to the two strict faces

$$a + b - c \geq j, \quad a + b - 2c \leq j - \eta_0. \quad (58)$$

## 6 Large-content pruning on long fibers

The  $+1$  in (43) is harmless when  $q = [r, s] \leq J$ . In that range we can sum the  $J/q$  occupancy and use row residue energy to remove every polynomially large common content.

For  $d \geq 1$ , define the absolute row congruence mass

$$\mathcal{R}_d := \sum_{\alpha, \gamma} |\gamma_\alpha^{(1)} \gamma_\gamma^{(2)}| \mathbf{1}_{d|m_\alpha - m_\gamma}. \quad (59)$$

**Lemma 6.1** (Row residue energy). *Uniformly for  $d \geq 1$ ,*

$$\boxed{\mathcal{R}_d \ll_{\mathcal{Q}} \left( \frac{Q^2}{d} + Q \right) \log(2L)}. \quad (60)$$

*Proof.* For  $i = 1, 2$  and  $a \pmod{d}$ , put

$$X_i(a) = \sum_{m_\alpha \equiv a \pmod{d}} |\gamma_\alpha^{(i)}|, \quad E_i(a) = \sum_{m_\alpha \equiv a \pmod{d}} |\gamma_\alpha^{(i)}|^2.$$

The distinct row integers lie in an interval of length  $O_{\mathcal{Q}}(Q)$ , so each residue class contains  $O_{\mathcal{Q}}(Q/d + 1)$  rows. Cauchy–Schwarz gives

$$X_i(a)^2 \ll_{\mathcal{Q}} \left( \frac{Q}{d} + 1 \right) E_i(a).$$

Hence

$$\begin{aligned} \mathcal{R}_d &= \sum_{a \pmod{d}} X_1(a) X_2(a) \\ &\ll_{\mathcal{Q}} \left( \frac{Q}{d} + 1 \right) \left( \sum_a E_1(a) \right)^{1/2} \left( \sum_a E_2(a) \right)^{1/2}. \end{aligned}$$

Apply (10). □

Let  $\mathcal{S}_{\leq Y, > D}$  denote any one of the three raw ultra channels, restricted to

$$[r, s] \leq Y \leq J, \quad (r, s) > D. \quad (61)$$

All original prefix and reflected-quotient support conditions remain in force.

**Theorem 6.2** (Long-fiber large-content pruning). *Uniformly for  $1 \leq D < Y \leq J$ ,*

$$|\mathcal{S}_{\leq Y, > D}| \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon \left\{ \frac{XQ}{D} \log^2(2Y) + X \log^3(2Y) \right\}. \quad (62)$$

*The same estimate holds for the sum of the three raw channels. If  $D \geq X^\eta$  for fixed  $\eta > 0$ , then*

$$|\mathcal{S}_{\leq Y, > D}| \ll_{\varepsilon, \eta, h, \mathcal{D}} XQX^{-\min\{\eta, \beta\} + \varepsilon}. \quad (63)$$

*Proof.* Put  $d = (r, s)$  and write

$$r = da, \quad s = db, \quad (a, b) = 1.$$

By target primitivity,  $(d, h) = 1$ ; by [lemma 4.1](#), compatibility therefore gives  $d \mid m_\alpha - m_\gamma$ . Also

$$[r, s] = dab.$$

For  $dab \leq Y \leq J$ , a compatible fiber has

$$O_{\mathcal{D}} \left( \frac{J}{dab} \right) \quad (64)$$

orbit points, with the constant absorbing the endpoint  $+1$ .

Take absolute values and absorb the divisor coefficients and fixed weights into  $X^\varepsilon$ . We retain the necessary condition  $d \mid m_\alpha - m_\gamma$  and the common interval support used in (64), but delete the channel-specific prefix, reflection, and squarefreeness selectors. For each exact  $d$ , the coprime parts contribute

$$\sum_{\substack{a, b \geq 1 \\ ab \leq Y/d}} \frac{1}{ab} \ll \log^2(2Y). \quad (65)$$

Thus [lemma 6.1](#) gives

$$\begin{aligned} |\mathcal{S}_{\leq Y, > D}| &\ll X^\varepsilon J \log^2(2Y) \sum_{D < d \leq Y} \frac{1}{d} \left( \frac{Q^2}{d} + Q \right) \\ &\ll X^\varepsilon \left\{ \frac{JQ^2}{D} \log^2(2Y) + JQ \log^3(2Y) \right\}. \end{aligned} \quad (66)$$

Use  $JQ \asymp X$  to obtain (62). If  $D \geq X^\eta$  and  $Q = X^{\beta+o(1)}$ , the two terms save respectively  $X^{-\eta}$  and  $X^{-\beta}$  relative to  $XQ$ ; logarithms and the  $o(1)$  factors are absorbed into  $X^\varepsilon$ .  $\square$

*Remark 6.3* (Two different content conditions). For  $q \leq J$ , the factor  $J/q$  makes the simple condition  $(r, s) > X^\eta$  sufficient. For  $q \geq J$ , the occupancy is only  $O(1)$  and a large gcd alone does not control the number of primitive pairs. The sparse theorem therefore requires the sharper condition  $q/(r, s) \leq JX^{-\eta}$ .

## 7 The high- $\beta$ ledger

We now place the two content theorems inside the exact parameter sample closed through  $T$  in TPC-28. The relevant exponents are

$$\beta = \frac{267}{400}, \quad j = 1 - \beta = \frac{133}{400}, \quad t = \frac{193}{500}. \quad (67)$$

Thus

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad T = X^{193/500+o(1)}. \quad (68)$$

The reflected complement variables satisfy

$$r, s \ll \frac{X}{T} = X^{307/500+o(1)}. \quad (69)$$

## 7.1 A sparse content-rich cell

Choose the rational block exponents

$$a = b = \frac{1}{4}, \quad c = \frac{1}{10}. \quad (70)$$

The lcm and primitive-lcm exponents are

$$a + b - c = \frac{2}{5}, \quad (71)$$

$$a + b - 2c = \frac{3}{10}. \quad (72)$$

Hence the sparse margin is

$$\frac{2}{5} - \frac{133}{400} = \boxed{\frac{27}{400}}, \quad (73)$$

while the incidence saving from (57) is

$$\eta_{\text{sp}} = \frac{133}{400} - \frac{3}{10} = \boxed{\frac{13}{400}}. \quad (74)$$

Both divisor scales  $X^{1/4}$  lie below the reflected upper range (69). In the both-ultra channel the two quotient divisors have size  $X^{3/4+o(1)} > T$ . In a mixed channel, the original prefix divisor  $X^{1/4}$  lies below  $T$ , while the reflected quotient on the other row again has size  $X^{3/4+o(1)}$ . Thus the same exponent cell is geometrically admissible in all three raw channels.

Write  $\mathcal{S}_{1/4,1/4,1/10}$  for the sum of the three raw-channel dyadic blocks with  $A = X^{1/4+o(1)}$ ,  $B = X^{1/4+o(1)}$ , and  $G = X^{1/10+o(1)}$ , with fixed dyadic endpoint constants.

**Corollary 7.1** (Exact high- $\beta$  sparse margin). *For every fixed  $\eta_0 < 13/400$ , the sum of the three physical raw channels over the block (70) satisfies*

$$\boxed{\mathcal{S}_{1/4,1/4,1/10} \ll_{\eta_0,h,\mathcal{D}} X Q X^{-\eta_0}}. \quad (75)$$

*Equivalently, the relaxed exponent ledger has exact margin  $13/400$ .*

*Proof.* Equations (71) and (72) verify the two faces in (58) with every fixed margin below  $13/400$ . Apply theorem 5.3. The  $o(1)$  scale factors are absorbed by the gap between  $\eta_0$  and  $13/400$ .  $\square$

*Remark 7.2* (Two different numbers). The TPC-28 full-rectangle conductor margin at its base was  $13/1600$ . The number  $13/400$  in (74) is a new and larger incidence margin for one reflected complement cell. The two fractions arise from different inequalities and must not be identified.

## 7.2 A long-fiber content cutoff

Take

$$D = X^{1/10}, \quad X^{1/10} < Y \leq J. \quad (76)$$

Since  $\beta = 267/400 > 1/10$ , theorem 6.2 gives, for every fixed  $\eta_1 < 1/10$ ,

$$\boxed{\mathcal{S}_{[r,s] \leq Y, (r,s) > X^{1/10}} \ll_{\eta_1,h,\mathcal{D}} X Q X^{-\eta_1}}. \quad (77)$$

For example, the scale  $r \asymp s \asymp X^{1/10}$  lies in the long range and has quotient divisors of size  $X^{9/10+o(1)} > T$ . Again, the theorem is an upper bound for an allowed region; it does not assert that every dyadic cell carries nonzero arithmetic mass.

Table 1: Exact exponent ledger at the TPC-28 high- $\beta$  sample.

| Quantity                            | Exact value | Role                   |
|-------------------------------------|-------------|------------------------|
| Orbit exponent $j$                  | 133/400     | natural fiber length   |
| Reflected upper exponent $1 - t$    | 307/500     | complement range       |
| Sparse lcm exponent $a + b - c$     | 2/5         | exceeds $j$            |
| Sparse margin                       | 27/400      | fiber is short         |
| Primitive-lcm exponent $a + b - 2c$ | 3/10        | content-rich cost      |
| Sparse incidence margin             | 13/400      | new fixed-power saving |
| Long content cutoff exponent        | 1/10        | long-fiber pruning     |

## 8 The primitive-core boundary and the next gate

The two theorems remove large shared content on both sides of the fiber-length transition. They do not remove the whole complement. This section records the exact remainder and prevents a sign-blind interpretation of the result.

### 8.1 The sparse method boundary

In exponent notation, a primitive complement pair has  $c = 0$ . The sparse condition (56) becomes

$$a + b \geq j, \quad (78)$$

whereas the incidence saving is

$$\eta_{\text{sp}}(a, b, 0) = (j - a - b)_+ = 0. \quad (79)$$

Thus theorem 5.1 reaches only the natural scale on the coprime sparse core. This is a sharp boundary of the present absolute-incidence argument, not a lower bound for the signed arithmetic sum.

For a fixed  $\eta > 0$ , after removing theorem 5.3, the remaining sparse region may be written as

$$[r, s] \geq J, \quad \frac{[r, s]}{(r, s)} > JX^{-\eta}. \quad (80)$$

It contains every coprime pair with  $rs \geq J$ . Although each such fiber has  $O(1)$  orbit points, the family has enough modulus pairs to restore the natural scale. A short fiber by itself is therefore not a cancellation theorem.

### 8.2 The remaining long fibers

For  $[r, s] \leq J$ , theorem 6.2 removes  $(r, s) > X^\eta$  for any fixed  $\eta > 0$ . What remains has

$$[r, s] \leq J, \quad (r, s) \leq X^\eta. \quad (81)$$

On any further subpacket carrying the additional generic row separation  $|m - n| \geq QX^{-\eta}$ , the affine determinant in (35) satisfies

$$\left| \frac{h(m - n)}{(r, s)} \right| \gg_h QX^{-2\eta}. \quad (82)$$

Thus the separated subpacket is not a small-determinant degeneracy. This conditional observation is not used in either closure theorem and does not remove any unseparated row pairs. The remaining long problem consists of small-content, polynomial-slope one- and two-point reflected

Möbius families: one quotient Möbius factor moves in a mixed channel, and two move in the both-ultra channel.

For a fixed compatible quadruple  $(m, n, r, s)$  in the both-ultra channel, the logarithmically averaged two-point Elliott theorem is a fixed-form cancellation benchmark for the two affine quotient forms in (33)–(34) [1]. A mixed channel has only a one-point moving Möbius factor. Neither fixed-form benchmark supplies an effective rate uniform over the polynomially growing slopes, the divisor family, and the physical row weights. It therefore cannot simply be summed over (81).

### 8.3 A sufficient next interface

The next useful statement is a family estimate, not a pointwise asymptotic for an  $O(1)$ -point fiber. One possible interface is the following.

**Definition 8.1** (Primitive reflected-family dispersion). For a fixed content threshold  $X^\eta$ , let  $\mathcal{P}_\eta$  be the union of the remaining regions (80) and (81), with the actual mixed and both-ultra coefficient types retained. A primitive reflected-family dispersion estimate is a bound

$$\sum_{\text{three raw channels}} \mathcal{S}(\mathcal{P}_\eta) \ll_{B,\eta,h,\mathscr{D}} XQ(\log X)^{-B} \quad (83)$$

for every fixed  $B > 0$ , uniformly over the physical opened-row weights and row-only mask.

Equation (83) is not assumed or proved in this paper. Its formulation preserves the joint cutoff difference: one must not first assert that each complete mixed channel or the complete both-ultra channel is negligible. The content-rich pieces may be removed separately only because the present theorems bound them absolutely.

### 8.4 Status after this paper

| Object                                                                               | Status                                                          |
|--------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Selected calibrated square through $T$                                               | Closed in TPC-28 under its physical conductor interfaces.       |
| Two row-drift legs beyond $T$                                                        | Fixed-power small by <a href="#">proposition 3.2</a> .          |
| Long fibers $[r, s] \leq J$ with $(r, s) > X^\eta$                                   | Closed with fixed-power saving by <a href="#">theorem 6.2</a> . |
| Sparse fibers with $[r, s] \geq J$ and $[r, s]/(r, s) \leq JX^{-\eta}$               | Closed with fixed-power saving by <a href="#">theorem 5.3</a> . |
| Small-content long fibers and primitive sparse fibers                                | Open; require signed family dispersion such as (83).            |
| Complete TPC-18 residual and endpoint reassembly                                     | Not proved.                                                     |
| Positivity, Hardy–Littlewood asymptotic, twin primes, or general parity breakthrough | Not claimed and not implied.                                    |

This is a genuine reduction of the open complement geometry: the remaining long core has small gcd, whereas the remaining sparse core has large primitive lcm  $[r, s]/(r, s)$ . Neither remaining condition is a signed cancellation theorem, and this is not a solution of the parity-sensitive core.

## 9 Exact certificate, dependencies, and scope

### 9.1 Finite exact certificate

The standard-library script

`experiments/tpc29_certificate.py`

uses exception-based checks in both ordinary and optimized Python. It verifies the finite algebra underlying the paper by independent direct and structured computations:

- (i) generalized-CRT compatibility, uniqueness of the orbit class, and finite-window occupancy for primitive row and modulus pairs;
- (ii) direct triple enumeration versus CRT enumeration for the dyadic incidence count, together with an explicit-constant finite version of (39);
- (iii) direct versus CRT long-fiber bookkeeping, including the exact factorization  $[r, s] = d(r/d)(s/d)$  for  $d = (r, s)$ ;
- (iv) the calibrated prefix identity (23) and the five-term joint identity (24), using formal prime-log vectors rather than floating-point logarithms;
- (v) the rational exponent formula (57), its boundary cases, and the exact high- $\beta$  margins in (73)–(74);
- (vi) explicit negative scope flags for every prime-pair, positivity, full-residual, and parity claim not proved here.

The incidence regression checks a finite inequality with its explicit arithmetic multiplicity constant. It does not assign constant one to the asymptotic  $\ll$  symbol. Nor does finite enumeration prove an asymptotic Möbius estimate. Ordinary and optimized Python both complete 37,265 recorded top-level exact checks and regenerate byte-identical JSON. The reproducibility values are

- source SHA-256: 13de31994c18d36f30975121daf485d8aca2f738f5a48dfc31e297130ed136c6;
- JSON SHA-256: 0cbe83d578f39246542d6b0480db3d4a3d9b02ed23036dd9f569671dd5e2242d;
- certificate digest: 5c2858662e76278cf3e8b4f1e26a4ac024e3b863ca21a3be8866dcf921a162b1.

The source hash normalizes newlines to LF; the final value hashes the canonical sorted payload before the digest field is inserted.

## 9.2 Dependency ledger

| Input or statement                                 | Status                              | Use                                                                             |
|----------------------------------------------------|-------------------------------------|---------------------------------------------------------------------------------|
| Opened-row $\ell^1/\ell^2$ provenance              | Imported                            | TPC-25; physical pair mass and row residue energy.                              |
| Calibrated prefix and $L^{-1}$ drift               | Imported<br>definition and<br>bound | TPC-28; exact cutoff difference used here.                                      |
| Selected truncated closure at $T$                  | Imported                            | TPC-28; locates the new complement, not used to prove the incidence bounds.     |
| Cutoff difference, CRT fiber, incidence theorem    | Proved here                         | Finite identity and elementary divisor-incidence geometry.                      |
| Sparse content-rich and long large-content closure | Proved here                         | Absolute physical estimates; no Möbius cancellation input.                      |
| Fixed-form logarithmic Elliott theorem             | External<br>benchmark               | Explains the fixed-fiber result but is not used in either main closure theorem. |
| Primitive reflected-family dispersion              | Open                                | Required for the remaining small-content signed core.                           |

## 9.3 Claims deliberately excluded

The following statements are not consequences of this paper:

- closure of every sparse fiber or every long fiber;
- closure of the complete ultra-long cutoff difference;
- closure of the full TPC-18 residual or every endpoint packet;
- a positive lower bound, a Hardy–Littlewood asymptotic, or infinitely many twin primes;
- a general violation of the sieve parity barrier;
- polynomially uniform two-point Chowla or Elliott cancellation.

The positive conclusion is narrower and exact: large shared content closes a complete long-fiber sector and a nonempty sparse incidence wedge in exponent space. The remaining core is explicitly identified rather than treated as random.



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